TAFE-CLIP: A Multimodal Model for Long-Tailed Remote Sensing Image Classification

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***Abstract*—Most existing deep learning models degrade sharply on a case, named "long-tailed remote sensing image classification," where tail classes with scarce samples are often overwhelmed by head categories, leading to persistent misclassification. To address this challenge, this article proposes a multimodal model, called "tail-aware feature enhancement contrastive language-image pretraining (TAFE-CLIP)." The model uses multimodal contrastive learning to align language and image information; by incorporating convolutional enhancement and channel attention modules, the model can extract local spatial features and further refine feature representations. The proposed TAFE-CLIP was validated on four standard remote sensing datasets: NWPU-RESISC45, PatternNet, RSSDIVCS, and RSI-CB256. With experimental results, it can be demonstrated that the proposed TAFE-CLIP model can reach high accuracy of 90.5%, 98.8%, 91.7%, and 97.1% for the long-tailed image classification of four datasets, respectively, representing average improvements of 27.5%, 9.2%, 12.1%, and 14.0% for four datasets relative to the eight most current commonly used deep learning models, respectively. These results confirm that the proposed TAFE-CLIP can effectively alleviate the long-tailed distribution in remote sensing image classification.**

***Index Terms*—Long-tailed distribution, remote sensing image classification, vision-language model.**

I. Introduction

THE advancement in remote sensing image classification with deep learning has largely improved our understanding from the pixel to scene level [1], [2], [3], [4], [5]. However, natural land cover distributions are inherently imbalanced, such as the long-tailed trait, which is universal in all remote sensing images [6], [7], [8]. Fig. 1 reveals that an abundance of samples is concentrated in head classes (20%), but there is a severe scarcity of data in tail classes (40%). Due to such distribution differences, previous researchers have focused more on training the head categories during the training process, neglecting or even ignoring the tail categories. This imbalance training mode significantly impacts the image classification accuracy. This implies that these early deep learning models, which neglect tail classes, have poor performance for tail-class classification [9], [10], [11], [12].

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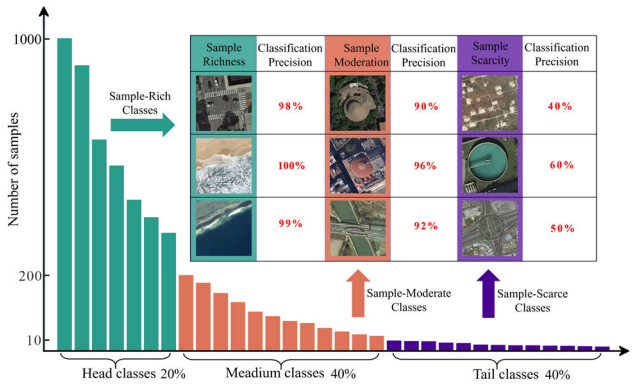


Fig. 1. Long-tailed class distribution diagram. The histogram shows the long-tailed distribution of samples across head (green), medium (orange), and tail (purple) classes.

This article proposes a method for long-tail remote sensing scene classification, which is named tail-aware feature enhancement contrastive language-image pretraining (TAFE-CLIP). The core contributions from this study are summarized as follows.

1) Leverage the visual-language contrastive learning mechanism of the CLIP model within a multimodal learning framework by combining image and text features for joint learning to better address the long-tail distribution problem.
2) Propose the TAFE module to explicitly resolve the feature ambiguity issue for tail categories in imbalanced long-tailed data. By combining a convolution enhancement (CE) unit with a squeeze-excitation (SE) module effectively captures the subtle features hidden in tail categories, thereby greatly enhancing the classification accuracy of tail classes.
3) Compare the proposed method with the eight most current commonly used deep learning models, which demonstrate that the proposed model has better classification performance in classifying the head, middle, and tail categories across four datasets, with a significant improvement in tail-class classification accuracy.

II. Related Work

The related work for remotely sensing image classification with the long-tailed distribution falls into three categories as follows.

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1) Resampling methods, which attempt to create balanced class distributions by modifying the training data through oversampling or undersampling. Oversampling augmented the sample counts of tail classes to boost their classification accuracy, allowed the model to prioritize underrepresented categories [13], [14], [15]. Alternatively, undersampling cut down the sample count of head classes to prevent the model from becoming overfitted to them [16], [17]. However, merely applying undersampling to head categories or oversampling of tail classes without incorporating novel feature information may lead to underfitting, thus weakening the model's generalization ability.
2) Loss reweighting methods, which adjust the formulation of loss functions via weight redistribution for tail classes, aim to boost learning emphasis on tail categories. A representative study in this area introduced focal loss, which added a modulation factor to reduce excessive attention to easily classified samples, thereby enhancing the performance for tail classes [18]. Cui et al. [19] proposed class-balanced (CB) loss, which adapted the weights of each class depending on its effective sample size to ensure tail classes were not ignored. By enhancing context-aware adaptive loss functions with multiscale features, Zhang et al. [20] proposed a gradient method that dynamically adjusted gradients during training, improving tail classes performance. Zhou et al. [21], [22] proposed a stable, efficient and high-precision approach to implement small object recognition within remote sensing images. Recently, the gradient-based dynamic reweighting methods have been applied to long-tailed classification tasks. Similarly, Ren et al. [23] proposed a balanced SoftMax function aimed at mitigating the dominance of head samples, which significantly enhanced the performance of tail categories.
3) Transfer learning methods. They served as a typical solution to alleviate the long-tailed distribution issue in remote sensing image classification tasks. They enhanced the generalization ability of small-sample tail classes by transferring the feature knowledge acquired from head classes [24], [25], [26]. Kang et. [27] decoupled the learning of feature representations from classifier training, achieving improved tail-class accuracy while sustaining the performance of head classes. In addition, few-shot learning techniques have been widely applied in long-tailed scenarios, where general feature representations are learned from abundant head-class samples and transferred to the tail categories [28]. Methods, such as prototypical networks [29] and meta-learning [30], demonstrated excellent performance in few-shot scenarios. To solve the overfitting issue in the primary models, Shi et al. [31] developed the LIFT approach, which focused on lightweight parameter adaptation schemes to mitigate the adverse impacts of over-fine-tuning on long-tailed category distribution. Emerging advances in vision-language models, represented by the CLIP architecture, have inspired new research directions. Although they do not directly target long-tail problems, they provide valuable insights and indirect solutions adaptable to these challenges. These can be subclassed as follows.
    a) CLIP, with the continuous progress of vision language-related technology [32], a variety of innovative model architectures have been proposed. In relevant research, MaskCLIP [33] merged masked self-distillation with vision-language contrastive learning to improve the transfer capability of visual encoders. A-CLIP [34] used attention masking to enhance training efficiency. FLIP [35] adopted masking strategies to reduce computational complexity and accelerate pretraining, without special optimization for long-tailed distributions. Relying on the powerful semantic modeling ability of large-scale pretraining, various CLIP based models have been proposed to address remote sensing tasks. RemoteCLIP [36] and GeoRSCLIP [37] were fine-tuned on large-scale remote sensing datasets, which can effectively improve the accuracy of cross-modal retrieval and zero-shot classification tasks and lay a solid foundation for imbalanced remote sensing data. Other research, GRAFT [38] and ChangeCLIP [39], optimized the cross-modal alignment strategy to provide a scheme for unsupervised remote sensing tasks without text annotation and adapt to long-tail classification. APPLeNet [40] and PIR-CLIP [41] show strong cross-domain generalization, suggesting their effectiveness in handling data imbalance. Additionally, Yuan et al. [42] and Li et al. [43] developed PE-RSITR and RS-MetaNet, respectively. Although all kinds of remote sensing models based on CLIP have been verified to have good performance, there is little research on the optimization of visual encoder for the characteristics of small sample features, which is the core research starting point of this article. This article proposes a multimodal long-tailed remote sensing classification framework named TAFE-CLIP. Based on CLIP's image–text contrastive learning, the framework constructs cross-modal semantic correlations. The TAFE module integrates SE channel attention and dual-branch CE to simultaneously model global channel dependencies and local fine-grained textures of tail classes. Without sacrificing the recognition accuracy of head and medium classes, the proposed method remarkably boosts the identification precision of scarce tail ground objects.
    b) Convolutional neural networks (CNNs), with superior spatial feature mining capability, have gained widespread adoption in remote sensing image classification tasks. Classic backbone networks, such as ResNet [12], DenseNet [46], and their derived models, were often used to extract texture, structure, and multiscale detail features from remote sensing images. At present, most of the mainstream solutions rely on sample resampling and weighted

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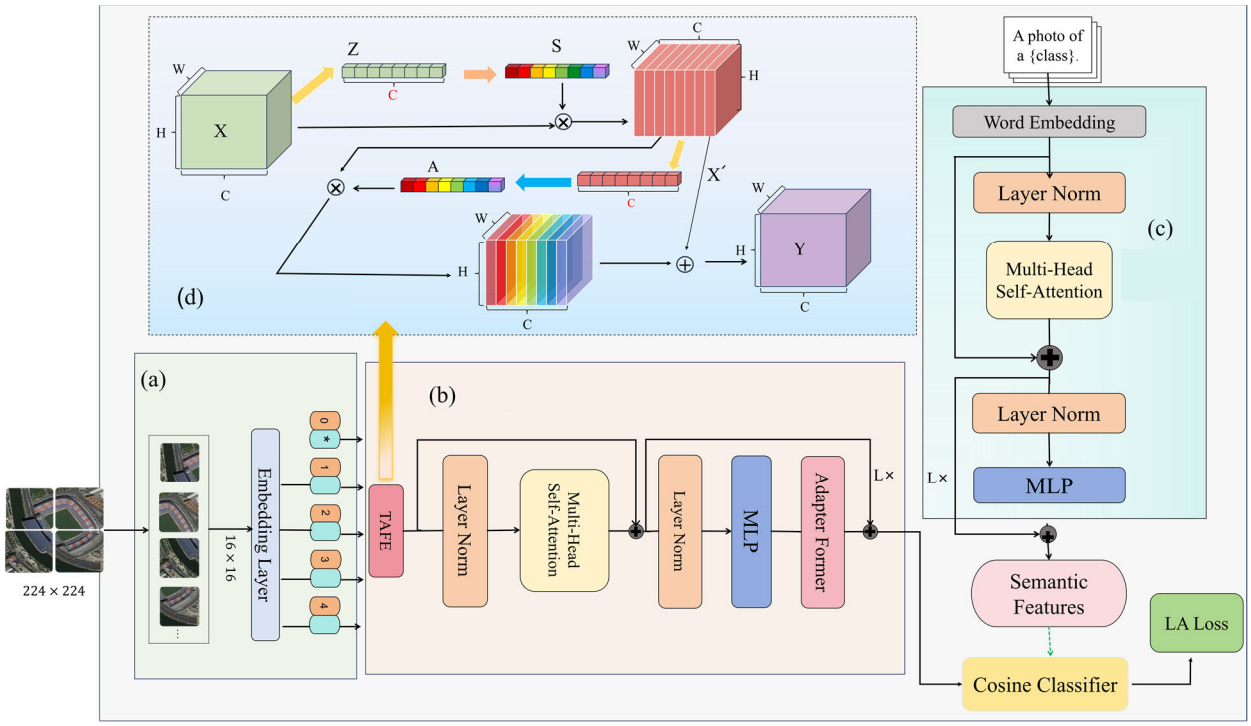


Fig. 2. Overall framework of TAFE-CLIP. The model consists of (a) text encoder generating semantic features, (b) image encoder processing images through patch embedding, TAFE attention mechanisms, (c) text encoder processing descriptions through word embedding and similar layers, with image and text features combined in a matrix to calculate similarity, and (d) TAFE module.

loss functions to alleviate the problem of category imbalance. However, this kind of traditional idea does not introduce the vision-language cross-modal matching strategy. At the same time, it lacks effective structural channel optimization means, thus failing to resolve feature ambiguity of tail classes in RS images.

c) SE [45] networks are widely adopted in remote sensing research, which can effectively make up for the lack of feature expression ability of traditional convolutional neural network in processing complex remote sensing scene data. RCA-Net [47] integrated the SE module into residual CNNs to enhance critical channels for scene classification. However, it lacked targeted optimization for the prevalent long-tailed distribution in RS datasets, resulting in poor rare-class recognition performance. The enhanced attention mechanisms, such as ECA [48] and GSoP [49], further improved the global modeling and small object perception ability of remote sensing images. Although convolutional operations and SE attention modules have been extensively applied in remote sensing, most SE-based methods in this field served as independent plug-ins for single-modal CNNs or transformers.

However, the existing methods have notable limitations. Resampling balances data but introduces noise without improving feature extraction. Remote sensing CLIP leverages pretrained knowledge yet lacks specialized structures for fine-grained remote sensing objects. Differing from the existing design ideas, this article fully integrates the local perception advantage of convolution operation and the adaptive correction ability of SE attention mechanism and proposes an integrated TAFE module. After embedding this module into CLIP visual encoder, it can extract multiscale information about ground objects, thereby addressing the issue of imbalanced sample distribution in long-tail classification tasks in remote sensing.

III. Method

A. Overall Framework of TAFE-CLIP

The overall structure of the TAFE-CLIP framework is illustrated in Fig. 2. It achieves cross-modal alignment between remote sensing image and text features via vision-language contrastive learning, which is mainly composed of two core modules: image encoder and text encoder. As shown in Fig. 2(c), the text encoder introduces prompt learning, integrates remote sensing scene category information into the text description, and generates text features that fit the features of the features. From Fig. 2(a), the image encoder uses the vision transformer (ViT) as the backbone network to segment the high-resolution remote sensing image into fixed-size patches and complete the feature mapping through the embedded layer. The image encoder is embedded with a TAFE module, which integrates the channel attention mechanism and CE strategies to synchronously improve the encoder's perception of local texture features and global semantic information of small-scale long-tail objects. As shown in Fig. 2, the cosine classifier is used to classify the aligned features and dynamically increase

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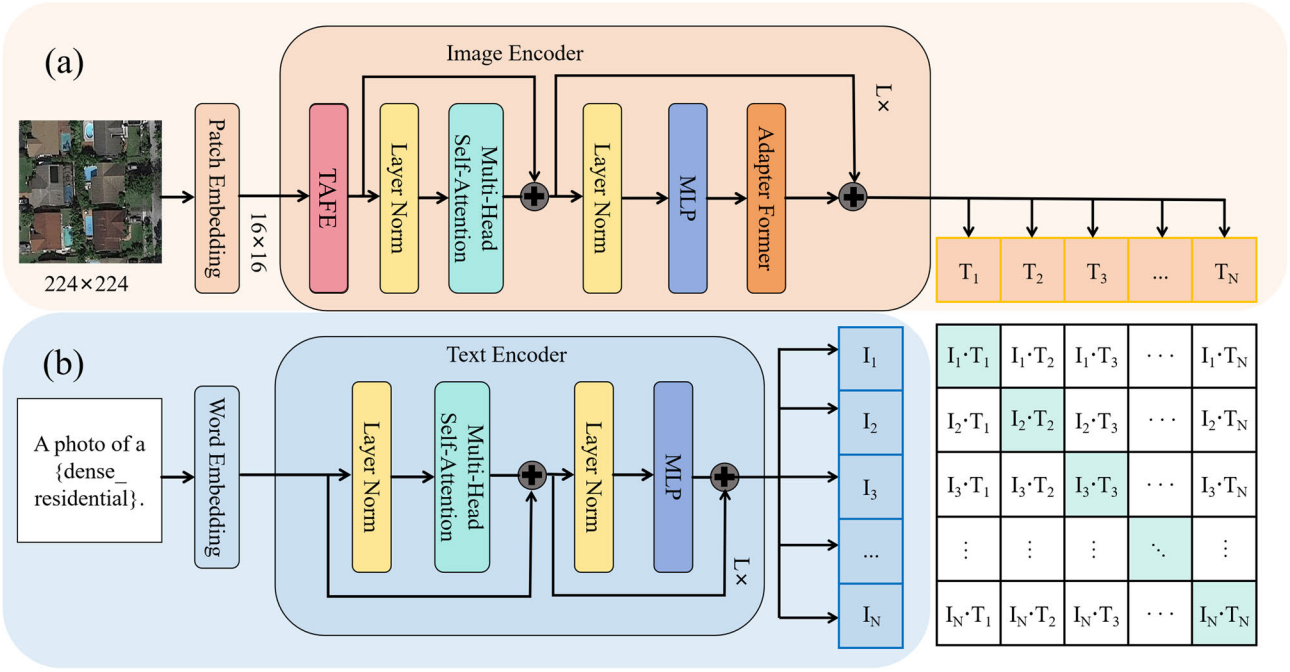


Fig. 3. TAFE-CLIP image and text encoder architecture. Illustrating (a) image encoder processing images through patch embedding, TAFE, and attention mechanisms and (b) text encoder processing descriptions through word embedding and similar layers, with image and text features combined in a matrix to calculate similarity.

the weight proportion of the long-tail feature categories in the model training process through the loss function.

B. Development of Image and Text Feature Extraction Model in TAFE-CLIP

In remote sensing image classification tasks with long-tail distribution, the tail classes often have insufficient samples and low feature discriminability, which restricts the recognition effect of the traditional classification model. To solve this problem, the TAFE module was designed for the core problem of remote sensing image long-tail task, that is, the lack of feature saliency of the tail classes and the loss of local detail information, and constructs a targeted feature enhancement mechanism to improve the feature expression ability of the tail classes. With the help of cross-modal comparative learning, the presented TAFE-CLIP model can effectively merge the visual features of remote sensing images and the semantic information of corresponding text descriptions.

As shown in Fig. 3, the image encoder selects ViT-B/16 as the backbone network. For remote sensing images, the model first divides the input images into 16 × 16 patches (matching the conventional scale of aerial features) by block embedding operation and linearly projects them into a high-dimensional space to produce initial features. These features are processed by the TAFE module, which combines the channel attention mechanism and CE method to enhance the local texture features of small-scale long-tail features while preserving the global semantic information. After global pooling, channel weight calibration, and convolution optimization, the features are input into the subsequent transformer layer. The residual connection was introduced after the self-attention module and the multilayer perceptron module to guarantee the stable transmission of gradient and improve the robustness of deep features. To further improve adaptability and flexibility, an Adapter Former [44] was introduced after the MLP. This Adapter Former refines feature processing, enhancing generalization capabilities without significantly increasing computational costs. Using this hierarchical encoding method, the TAFE-CLIP image encoder processes image inputs to generate feature vectors T_i (i = 1, 2,..., K), global semantic information, and local spatial features of the remote sensing images.

The text encoder is illustrated in Fig. 3(a). Based on the standard transformer architecture, it generates text embeddings for remote sensing ground object categories. The textual input is generated through prompt learning, where N category labels from a transfer dataset are combined with a prompt template, forming an input format like "a photograph of a {class}," where {class} is replaced by category names, such as "dense_residential." This input format helps the model distinguish between categories. Initially, the text input is processed through an embedding layer, mapping each word into a high-dimensional feature space. Layer normalization combined with multihead self-attention mechanisms associates various parts of the text. After passing through residual connections and layer normalization, the attention output enters an MLP for further feature extraction. Finally, by stacking multiple encoder layers, the text input generates feature vectors I_j (j = 1, 2,..., K), providing a rich textual representation for each category.

After processing by the image and text encoders, the resulting image feature vectors T_i (i = 1, 2,..., K) and text feature vectors I_j (j = 1, 2,..., K) are compared using cosine similarity,

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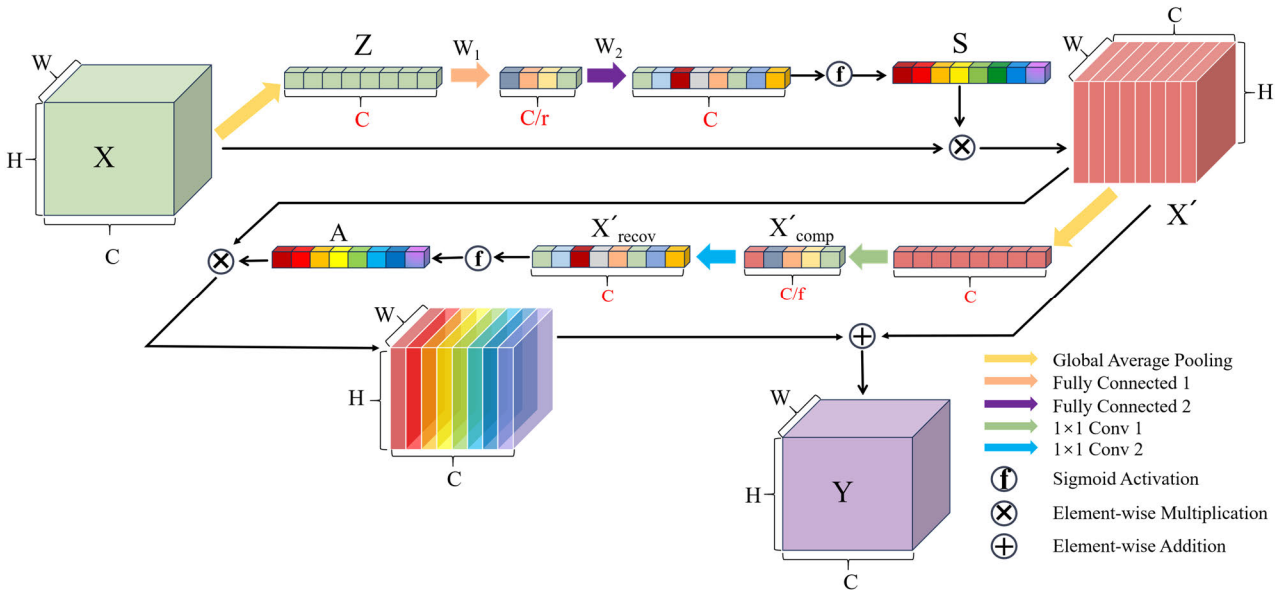


Fig. 4. Architecture of the TAFE module. It contains an SE channel attention branch for global channel calibration and a dual 1 × 1 CE branch for fine-grained local feature extraction.

as shown in the matrix in Fig. 3. By calculating these similarities, the model matches the image and text features, thereby selecting the most semantically similar category to complete image classification.

C. Development of Convolution-Enhanced and SE TAFE

Classic SE modules modeled global channel features via global average pooling and then used two fully connected layers to compress and reconstruct channel dimensions. Finally, sigmoid activation generates adaptive channel weights. However, in the task of remote sensing long-tail classification, tail objects generally have the problem of extreme lack of samples, which leads to insufficient feature learning. Traditional SE modules struggle to effectively distinguish subtle visual differences within tail categories (such as the subtle distinctions between "artificial grass" and "green farmland"), which can easily result in category confusion.

The TAFE module proposed in this article is the core feature extraction module of a multimodal framework. This module was improved based on the classic SE attention mechanism [45] and improved the ability of feature expression. Fig. 4 depicts the structure schematic of the TAFE module. The module integrates channel attention mechanism and CE unit to realize the complementary aggregation of global semantic information and local texture features. By embedding convolution operations into the channel-weighting process, the TAFE module effectively captures fine-grained spatial dependencies in image features, thereby boosting recognition performance for minority samples in long-tailed distribution of remote sensing images.

Channel Attention Optimization: This mechanism preserves the SE module's ability to learn global channel features and generates channel-level weights through global average pooling, enabling the model to concentrate on feature channels that are critical for recognizing tail-end classes.

Local Feature Enhancement Design: After channel weighting, two 1 × 1 convolutional layers are introduced to enhance local features:

The first 1 × 1 convolution reduces the channel dimension to C/f, which not only preserves global contextual cues but also lowers computational overhead and further concentrates on the low-dimensional feature distribution of tail classes.

The second 1 × 1 convolution returns channel dimension to the original dimension C, capturing local spatial dependencies through convolution.

Local spatial attention weights A_c are generated using a sigmoid activation function, which enables the network to strengthen essential local features in tail-class samples.

A detailed account of each processing step involved in the TAFE module is provided in Sections III-D and III-E, including the channel-weighting mechanism and convolution-enhanced feature extraction methods.

D. SE Attention

To solve the insufficient tail-class representation problem caused by the long-tail distribution of remote sensing images, the TAFE module employs an SE mechanism in the initial stage to enhance features for tail classes by mining intrinsic correlations among feature channels. As displayed in Fig. 4, the workflow begins with the top-left input tensor X. It then proceeds through three key steps: global average pooling, channel compression and expansion, and channel reweighting. These steps culminate in the output, which achieves feature optimization and enhancement.

The first step (1) adopts global average pooling to compress spatial details across all input feature map channels, yielding the global feature representation Z [50]

$$z_c = \frac{1}{H \times W} \sum_{i=1}^{H} \sum_{j=1}^{W} X_c(i, j) \quad \forall c \in \{1, 2, \ldots, C\} \tag{1}$$

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where Z_c represents the global feature extracted by channel C, C means the total channel number, and H and W denote the spatial height and width of the feature map, respectively.

Next, global feature Z passes through two fully connected layers to achieve channel compression and dimension expansion. The first layer reduces the channel count to C/r with a ReLU activation to capture nonlinear interactions between channels. The second layer then reinstates the original channel count C and produces channel attention weights S using the sigmoid activation function [45]

$$s = \sigma\big(W_2\delta\left(W_1 z\right)\big) \tag{2}$$

where $W_1 \in R^{C/\mathrm{r} \times C}$ and $W_2 \in R^{C \times C/r}$ are the weight matrices for channel compression and expansion, respectively, δ represents the ReLU activation function, and δ corresponds to the sigmoid function.

Ultimately, the elementwise multiplication is utilized to apply the calculated channel weights S to the feature map of input data X, thereby tuning the significance of features in each channel. This improves the representation of critical channels while suppressing irrelevant activations, effectively completing the e-weighting of the input features [45]

$$X_{c'} = X_c \cdot s_c \quad \forall c \in \{1, 2, \ldots, C\}. \tag{3}$$

This dynamic channel weighting scheme can adaptively adjust the contribution of each channel to the final output, increasing the network's sensitivity to global features while also preserving and reinforcing critical features.

E. Development of Convolution-Enhanced Method

To capture fine-grained local texture of tail-class ground objects, we add a CE branch after the SE channel attention path. Fig. 4 shows the convolutional enhancement process, which includes two 1×1 convolution layers. The two-layer convolution completes the channel dimension compression and feature graph dimension restoration in turn and finally obtains the spatial attention weight.

The feature obtained after processing by the SE module undergoes channel compression through the first 1×1 convolution layer to implement channel compression, thus reducing the original channel count C from its initial value to C/f (4). This effectively reduces computational complexity while preserving the global contextual information extracted through global pooling. Next, the second 1×1 convolutional layer restores the number of channels to the initial dimension C (5), thereby ensuring that the output feature map matches the input in channel dimension. This process captures and integrates fine-grained local features within the low-dimensional representation

$$X'_{\mathrm{comp}} = W_{comp} * X' \tag{4}$$

$$X'_{recov} = W_{recov} * X'_{comp} \tag{5}$$

where $*$ represents the convolution operation, f is the compression factor, $W_{comp} \in R^{C/f \times C \times 1 \times 1}$ is the weight of the first 1×1 convolution kernel, and $W_{recov} \in R^{C \times C/f \times 1 \times 1}$ is the weight of the second 1×1 convolution kernel.

Through two convolution-based linear transformations, the TAFE module not only retains global feature information from channel weighting but also enhances the interaction and saliency of local spatial features. Furthermore, the feature map processed by the convolutions generates local spatial attention weights A_c is generated using a sigmoid activation function (6), enabling fine-grained weight adjustment

$$A_{c'} = \sigma\left(X_{cov'}\right) \quad \forall c \in \{1, 2, \ldots, C\}. \tag{6}$$

The generated attention weight A_c represents the importance of local spatial features and performs a fine-grained weight adjustment at every spatial position within the feature map, which is applied to subsequent weighting operations.

Weighted Fusion and Residual Connections: In TAFE, we multiply convolution-generated spatial weights with channel attention weights elementwise and then fuse the result with the original input via residual connection. To preserve effective feature information, the TAFE module leverages residual shortcuts to directly aggregate original input features into refined output features

$$Y_{c'} = A_{c'} \cdot X_{c'} + X_{c'} \quad \forall c \in \{1, 2, \ldots, C\} \tag{7}$$

where $Y_{c'}$ is the fused discriminative feature vector corresponding to category c', $A_{c'}$ is a category-specific learnable matrix for fine-grained feature scaling, and $X_{c'}$ represents the raw high-resolution embeddings of category c' samples. The constraint $\forall$ c $\in$ {1, 2,...,C} means the fusion formula applies to all C.

F. Tail-Class Classification With Cosine Classifier and LAs

To address the inherent class imbalance in long-tail distribution datasets, this study employs an enhanced cosine classifier combined with a logit adjustment (LA) loss function. The cosine classifier normalizes both features and weights to eliminate magnitude discrepancies. Meanwhile, LA loss incorporates the prior distribution of per-class samples to amplify the loss gradients of tail classes.

Traditional linear classifiers are affected by class imbalances in long-tailed datasets, leading to high accuracy on head classes and inferior generalization on tail classes. This is because such classifiers are sensitive to the magnitude of class weight vectors. To eliminate this bias and construct a fair decision boundary for tail classes, the TAFE-CLIP model introduces a cosine classifier. TAFE-CLIP employs a cosine classifier that normalizes both the input feature vectors and the category weight vectors, ensuring that classification decisions depend solely on the direction of the vectors rather than their length. Formula (8) of the cosine classifier is defined as follows [50], [51]:

$$z_j = \sigma \cdot \frac{w_j^T \cdot f}{\left\|w_j\right\|_2 \cdot \left\|f\right\|_2} \tag{8}$$

where Z_j represents the predicted logit score for class j, w_j is the weight vector of class j, f represents the input feature vector, σ is the scaling factor, and T is the temperature scaling.

To improve the model's ability to accurately distinguish tail-class samples, this study also introduces a scaling

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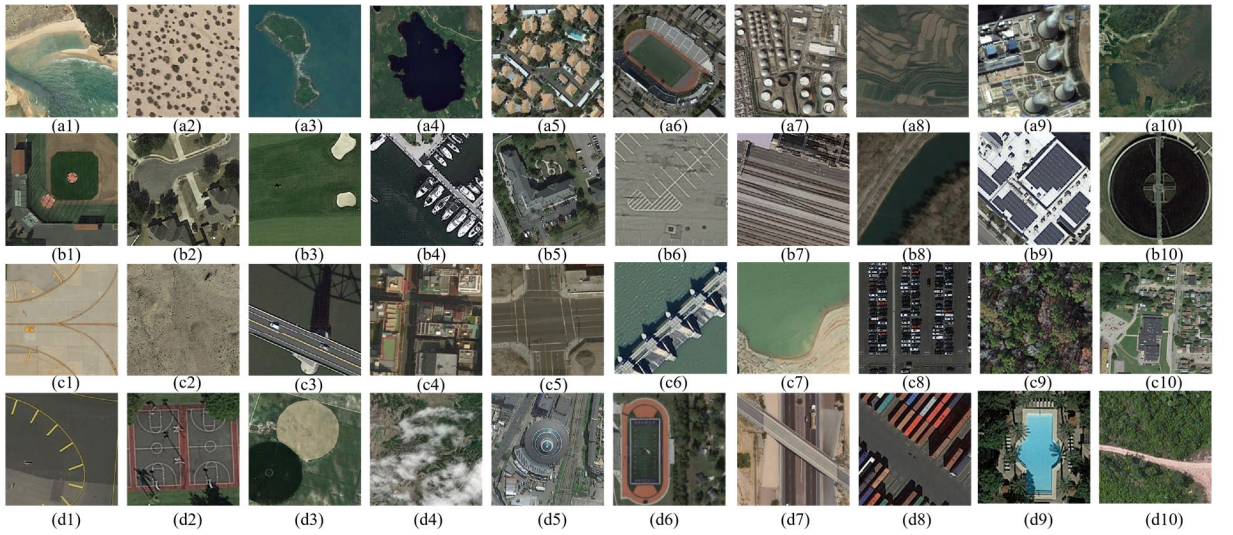


Fig. 5. Example scene images from each dataset.

factor σ and a temperature coefficient T into the cosine classifier to optimize interclass decision boundaries. The introduction of the scaling factor adjusts the interclass margin, whereas the temperature coefficient further reduces prediction bias by controlling class distribution smoothness. Problems optimization for long-tail problems: through normalization, the cosine classifier balances the decision boundaries across all classes, which is particularly beneficial for recognizing long-tail classes.

This experiment adopts LA loss to eliminate the model bias caused by imbalanced class distribution. To ensure that tail classes receive more attention during model training, the LA loss function is adopted, whose expression is defined as follows [52]:

$$L_A(x, y = j) = -\log \frac{\exp\left(z_j + \log P(y = j)\right)}{\sum_{k=1}^{K} \exp\left(z_k + \log P(y = k)\right)} \quad (9)$$

where $P(y = j)$ stands for the prior probability of class j calculated according to the category distribution within the training dataset, and K indicates the total class quantity.

IV. Experiment and Results

A. Datasets

We validated TAFE-CLIP on four publicly available remote sensing datasets NWPU-RESISC45 [3], PatternNet [53], RSSDIVCS [54], and RSI-CB256 [59]. The detailed parameters for the four datasets are shown in Table I.

In this experiment, the datasets were randomly split into training and test subsets in a 7:3 ratio. The training set was then modified to create a long-tailed distribution. The test set maintains the original balanced distribution of each category without any sample reduction to guarantee objective classification evaluation for scarce tail classes. After random shuffling of all categories, the samples were divided into head, medium, and tail subsets according to the long-tailed distribution settings. The head classes, comprising 20% of the categories, retained all their samples, whereas the middle classes, comprising 40% of the categories, randomly retained 50%–90% of their samples. The tail classes, also comprising 40% of the categories, retained only a minimal number of samples, with each category retaining only ten images. A random selection method was applied to select the categories and samples. Fig. 5 shows the sample images from the four datasets.

TABLE I
Summary of Dataset Attributes

Dataset	Number of Categories	Number of Images	Total Number of Samples	Image Size
NWPU-RESISC45	45	700	31500	256×256
PatternNet	38	800	30400	256×256
RSSDIVCS	70	800	56000	256×256
RSI-CB256	35	350-1300	>24000	256×256

1) The NWPU-RESISC45 dataset comprises 45 scene categories with 700 images apiece, covering human activities, natural landscapes, and infrastructure, such as airports, ports, beaches, rivers, and residential areas, consisting of 38 categories with balanced sample volumes. The data are shown in Table II.
2) The PatternNet dataset comprises 30 400 remote sensing images, where each category contains 800 images, representing scenes, such as basketball courts, harbors, and swimming pools. The data are shown in Table III.
3) The RSSDIVCS dataset integrates multiple remote sensing image scene datasets, with 70 categories and

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TABLE II
SPECIFIC INFORMATION REGARDING NWPU-RESISC45

Class name	Training Samples	Class name	Training Samples
Basketball court	489	Church	284
Beach	489	Thermal power station	275
Dense residential	489	Parking lot	257
Desert	489	Ship	252
Golf course	489	Baseball diamond	10
Harbor	489	Bridge	10
Lake	489	Chaparral	10
Rectangular farmland	489	Circular farmland	10
Tennis court	489	Forest	10
Mobile home park	424	Industrial area	10
Airport	417	Intersection	10
Runway	410	Island	10
Palace	392	Meadow	10
Mountain	387	Medium residential	10
River	384	Overpass	10
Stadium	362	Railway station	10
Railway	360	Roundabout	10
Commercial area	345	Sea ice	10
Terrace	326	Snow berg	10
Cloud	325	Sparse residential	10
Freeway	317	Storage tank	10
Airplane	302	Wetland	10
Ground track field	288		

TABLE III
SPECIFIC INFORMATION REGARDING PATTERNNET

Class name	Training Samples	Class name	Training Samples
Chaparral	560	Cemetery	317
Closed road	560	Shipping yard	312
Nursing home	560	Dense residential	291
River	560	Basketball court	10
Runway	560	Beach	10
Solar panel	560	Christmas tree farm	10
Swimming pool	560	Coastal mansion	10
Bridge	486	Football field	10
Parking space	450	Forest	10
Crosswalk	432	Freeway	10
Baseball field	431	Golf course	10
Harbor	423	Oil gas field	10
Railway	417	Oil well	10
Ferry terminal	410	Overpass	10
Parking lot	395	Sparse residential	10
Runway marking	390	Storage tank	10
Intersection	385	Tennis court	10
Mobile home park	356	Transformer station	10
Airplane	328	Wastewater Treatment plant	10

800 images per category. The data are shown in Table IV.

4) The RSI-CB256 dataset comprises 24 000 images organized into six main categories and 35 subcategories, including agricultural land, transportation, and water resources. The data are shown in Table V.

B. Experimental Environment and Evaluation Metrics

The experimental conditions are detailed in Table VI, including the hardware and software configurations used. For the model in this article, the SGD optimizer was utilized with a batch size of 128, the weight decay set to 5×10^{-4}, and a momentum of 0.9. All the datasets were trained for 60 epochs. Adapter Former's bottleneck dimension was set to yield fewer learnable parameters than the classifier. During training, we employed the LA loss and configured and the cosine classifier's scaling factor σ as 25.

TABLE IV
SPECIFIC INFORMATION REGARDING RSSDIVCS

Class name	Training Samples	Class name	Training Samples
Airport	560	Resort	320
Bare land	560	Oil well	317
Cemetery	560	Sea island	316
Crosswalk	560	Oil gas field	313
Freeway	560	Airplane	295
Mangrove	560	Circular farmland	291
Mobile home park	560	Industrial area	290
Overpass	560	Wetland	290
Railway station	560	Mountain	288
Roundabout	560	Airport runway marking	10
Sea ice	560	Baseball field	10
Storage tank	560	Basketball court	10
Swimming pool	560	Beach	10
Airport	560	Bridge	10
Bare land	560	Waste water plant	10
Cemetery	560	Christmas tree farm	10
Crosswalk	560	Commercial area	10
Tree lined path	560	Convention center	10
Viaduct hub	493	Dam	10
Dense residential	490	Desert chaparral	10
Transformer station	474	Harbor	10
Tennis court	456	Intersection	10
Thermal power station	456	Lake	10
Football field	449	Lakeshore	10
Grass	449	Palace	10
Ship	433	Park	10
School	422	Pipeline	10
Snow mountain	412	Railway	10
Terrace	403	Rectangular farmland	10
Sclerophyllous fores	400	River	10
Church	389	Sea	10
Parking space marking	385	Shipping yard	10
Golf course	360	Shrub wood	10
Parking lot	356	Solar panel	10
Cloud	355	Sparse residential	10
Desert	324	Square	10
Deadened road	322	Stadium	10

To comprehensively evaluate the accuracy of the designed model in long-tailed classification, we selected the following key metrics: overall accuracy, macro-averaged $F1$ score, head classes accuracy, middle classes accuracy, and tail classes accuracy [8]. These metrics fully captured both the model's overall performance and its performance differences across category types.

C. Comparison of Overall Accuracy and F1-Score for Different Models

For a comprehensive evaluation of the TAFE-CLIP model on long-tailed distribution datasets, we perform experiments on four open-source remote sensing datasets: NWPU-RESISC45, PatternNet, RSSDIVCS, and RSI-CB256. Our experiments involve benchmarking the proposed model with different classic deep learning-based methods (such as

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TABLE V
SPECIFIC INFORMATION REGARDING RSI-CB256

Class name	Training Samples	Class name	Training Samples
Desert	764	River	190
Sea	719	Bridge	167
Mountain	568	Pipeline	138
Forest	542	Artificial grassland	10
Container	411	City building	10
Avenue	380	Dam	10
Mangrove	378	Dry farm	10
Airport runway	376	Highway	10
Sand beach	375	Marina	10
Stream	362	Parking lot	10
Hirst	349	Residents	10
Coastline	321	River protection forest	10
Bare land	315	Sapling	10
Green farmlanc	278	Shrub wood	10
Lakeshore	254	Snow mountain	10
Airplane	219	Sparse forest	10
Crossroads	216	Storage room	10
Town	208		

TABLE VI
EXPERIMENTAL ENVIRONMENT

Environment	Versions or Model Number
CPU	AMD Ryzen 9 7950X
GPU	NVIDIA GeForce RTX 4090
OS	Ubuntu 22.04
CUDA	11.7
Python	3.10.9
Pytorch	2.0.0

ResNet-101, EfficientNet [55], ViT-L [56], and Swin Transformer [57]) and multimodal models (such as CLIP, with ResNet, ViT-B/16, and ViT-B/32 as the image encoder), RemoteCLIP, GeoRSCLIP. To address the long-tailed data processing issue, we used several commonly used effective loss functions, including the LA for long-tailed distribution (LADE) [58], CB loss, and focal loss, to further evaluate improvements in tail classes.

To present the performances of the various models on the four datasets (NWPU-RESISC45, PatternNet, RSSDIVCS, and RSI-CB256) more intuitively, the line charts of the overall accuracy (OA) and $F1$ scores were plotted (Fig. 6). As shown in the results, across all datasets, there was consistent evidence of a substantial performance gap between the TAFE-CLIP framework and other benchmark methods. Furthermore, the TAFE-CLIP model consistently outperforms the others from the perspective of OA and macro $F1$ score, achieving the highest scores among all compared models.

TAFE-CLIP performed better than other baselines on all four datasets, with the most significant improvement in tail-class classification. Tables VII–X list the overall accuracy (OA), macro-$F1$ score, and the classification accuracy of different competing models across head, middle, and tail classes splits.

For the NWPU-RESISC45 dataset (Table VII), the proposed TAFE-CLIP framework achieved superior performance with 93.2% OA and 93.1% $F1$-score. The second-best performance

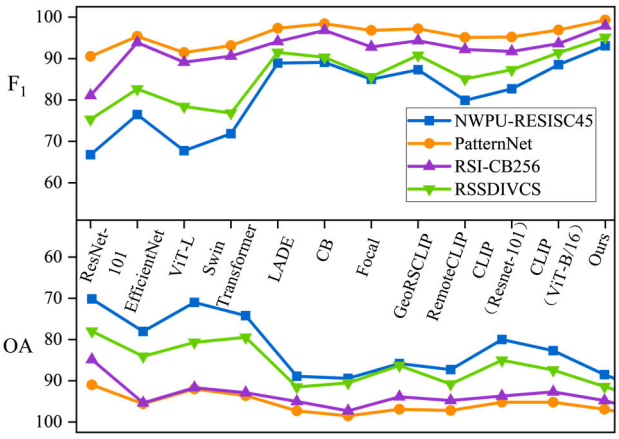


Fig. 6. Comparison of $F1$ score and OA for different models and methods across multiple datasets.

TABLE VII
ACCURACY COMPARISON OF VARIOUS MODELS ON THE NWPU-RESISC45 DATASET (%)

Model	OA (%)	F_1 (%)	Head (%)	Medium (%)	Tail (%)
ResNet-101	70.15	66.78	93.05	89.06	39.81
EfficientNet	78.02	76.48	94.37	92.23	55.66
ViT-L	70.98	67.74	94.63	90.62	39.52
Swin Transformer	74.2	71.87	95.1	91.4	46.6
LADE	88.9	88.9	89.8	88.7	88.3
CB	89.4	89.1	98.1	94.8	79.6
Focal	85.8	85.0	98.4	95.5	69.9
RemoteCLIP	88.6	88.6	92.2	89.2	87.7
GeoRSCLIP	87.3	87.3	88.4	86.8	88.1
CLIP(Resnet-101)	82.7	82.7	82.1	80.8	84.3
CLIP(ViT-B/16)	77.4	74.6	77.4	94.1	52.2
CLIP(ViT-B/32)	81.3	89.9	97.6	95.2	60.4
Ours	**93.2**	**93.1**	**96.7**	**94.0**	**90.5**

was by CB, attaining 89.4% OA and 89.1% in terms of $F1$ metric. Despite the improvement shown by CB, there was still a gap compared with TAFE-CLIP. LADE achieved an OA and $F1$ score of 88.9%, which were slightly lower than those of CB. Focal loss had an OA of 85.8% and an $F1$ score of 85.0%, performing below the other two. The above distinctions are prominently reflected in the line chart, where the TAFE-CLIP OA and $F1$ scores are positioned at the top, demonstrating their significant advantages in classification tasks.

In the PatternNet dataset (Table VIII), the TAFE-CLIP model achieved OA and $F1$ score of 99.3%, nearly achieving optimal performance for the classification task. The closest performance was from the CB, with an $F1$ score of 96.9% and an OA of 98.5%. Although CB performed well, it fell short of TAFE-CLIP. ResNet-101 and ViT-L had OA of 90.98% and 92.0%, correspondingly, with $F1$ values reaching 90.52% and 91.46%, respectively, which were significantly lower than TAFE-CLIP. The line chart clearly shows that the TAFE-CLIP is at the highest point on this dataset, far ahead of the other models.

In the RSI-CB256 dataset (Table IX), the TAFE-CLIP model reached an overall accuracy of 98.1% and $F1$ metric of 97.9%, consistently maintaining its leading performance

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TABLE VIII
ACCURACY COMPARISON OF VARIOUS MODELS ON THE PATTERNNET DATASET (%)

Model	OA (%)	F_1(%)	Head (%)	Medium (%)	Tail (%)
ResNet-101	90.98	90.52	99.17	98.69	79.06
EfficientNet	95.59	95.35	99.40	99.61	89.50
ViT-L	92.0	91.46	99.11	99.28	80.94
Swin Transformer	93.61	93.12	99.46	98.64	85.58
LADE	97.3	97.3	98.9	98.2	95.7
CB	98.5	98.4	99.3	99.8	96.8
Focal	96.9	96.8	99.9	99.8	92.8
RemoteCLIP	95.2	95.1	90.5	95.0	95.3
GeoRSCLIP	97.2	97.2	96.2	96.9	97.6
CLIP(Resnet-101)	95.2	95.2	91.9	95.4	96.3
CLIP(ViT-B/16)	92.4	92.1	99.3	99.3	82.9
CLIP(ViT-B/32)	96.4	96.4	99.6	99.6	92.1
Ours	**99.3**	**99.3**	**99.5**	**99.7**	**98.8**

TABLE IX
ACCURACY COMPARISON OF VARIOUS MODELS ON THE RSI-CB256 DATASET (%)

Model	OA (%)	F_1(%)	Head (%)	Medium (%)	Tail (%)
ResNet-101	84.85	81.08	98.43	95.88	63.51
EfficientNet	95.35	93.85	99.17	97.88	88.27
ViT-L	91.68	89.11	98.72	96.73	79.34
Swin Transformer	92.88	90.59	98.83	97.0	81.6
LADE	95.0	94.1	96.7	95.1	94.0
CB	97.3	96.8	99.1	98.3	94.6
Focal	93.9	92.8	98.4	98.9	85.6
RemoteCLIP	93.7	92.2	94.8	91.8	94.4
GeoRSCLIP	94.8	94.3	93.9	93.6	96.1
CLIP(Resnet-101)	92.7	91.7	93.6	90.4	93.0
CLIP(ViT-B/16)	83.8	83.3	98.8	97.5	66.0
CLIP(ViT-B/32)	95.0	93.9	99.0	99.0	88.0
Ours	**98.1**	**97.9**	**99.0**	**98.6**	**97.1**

among all experimental models. The CB model registered 97.3% in OA and 96.8% in terms of $F1$ performance, though it did not surpass TAFE-CLIP. LADE achieved 95.0% in OA performance and 94.1% in $F1$ evaluation, slightly below CB's performance. Focal loss recorded an OA of 93.9% and 92.8% $F1$ metric, markedly lower than CB and LADE. In comparison, ResNet-101 and ViT-L achieved OA scores of 84.85% and 91.68% and $F1$ scores of 81.08% and 89.11%, respectively, highlighting a significant gap with TAFE-CLIP. The line chart clearly indicates that TAFE-CLIP consistently performed the best on this dataset, demonstrating its strong dominance in classification tasks.

For the RSSDIVCS dataset (Table X), our proposed TAFE-CLIP achieves the best overall performance, securing 95.2% OA and a 95.1% $F1$ score. LADE came next by obtaining OA and $F1$ score of 91.5%, nevertheless, a notable discrepancy exists relative to our TAFE-CLIP. The CLIP(ViT-B/16) had an OA and $F1$ score of 91.4%, performing worse than the LADE. The line chart clearly shows that the TAFE-CLIP curve consistently remained at the highest point on this dataset, significantly outperforming the other models, indicating its clear advantage in complex classification tasks.

D. Tail Classification Accuracy Performance

Tail categories in long-tailed distribution datasets pose significant challenges for classification models due to sample scarcity. The TAFE-CLIP model's performance on tail categories was validated by comparing several categories from the four datasets. The radar chart (Fig. 7) illustrates performance differences among the models in these categories.

TABLE X
ACCURACY COMPARISON OF VARIOUS MODELS ON THE RSSDIVCS DATASET (%)

Model	OA (%)	F_1(%)	Head (%)	Medium (%)	Tail (%)
ResNet-101	77.97	75.31	77.32	85.88	70.39
EfficientNet	84.08	82.66	83.07	91.49	77.19
ViT-L	80.68	78.44	77.80	89.94	72.87
Swin Transformer	79.46	76.84	76.04	88.3	72.32
LADE	91.5	91.50	90.4	93.3	89.3
CB	90.5	90.3	93.9	95.8	83.6
Focal	86.3	85.6	97.0	97.3	70.0
RemoteCLIP	85.0	85.1	86.5	87.1	81.8
GeoRSCLIP	90.8	90.8	92.8	91.8	89.4
CLIP(Resnet-101)	87.4	87.3	88.0	88.7	85.9
CLIP(ViT-B/16)	79.1	77.0	95.8	96.7	52.5
CLIP(ViT-B/32)	87.3	86.2	98.5	98.5	70.7
Ours	**95.2**	**95.1**	**97.9**	**97.3**	**91.7**

In the NWPU-RESISC45 experiments, the global classification accuracy of the TAFE-CLIP model for the tail categories reached 90.5%, which was significantly higher than ResNet-101 (39.81%) and ViT-L (39.52%). For specific tail classification accuracy, such as wetland and medium residential, TAFE-CLIP achieved over 90% accuracy, whereas ResNet-101 and ViT-L achieved classification accuracy of less than 40% for these classification accuracies. As shown in the radar chart, TAFE-CLIP maintained a stable and high-precision classification across several tail classes (e.g., medium residential and island), whereas the curves of ResNet-101 and ViT-L exhibited considerable fluctuations, indicating significant shortcomings. Particularly in the snowberg category, the accuracy of TAFE-CLIP was perfect, whereas the performances of the other models were highly unstable.

In the PatternNet dataset, the proposed TAFE-CLIP attained a classification accuracy of 98.8% for the tail classification accuracy. For example, in the parking lot and soccer field category identification tasks, the categorization accuracy of TAFE-CLIP was close to 99%, whereas those of ResNet-101 and ViT-L were around 80% and 79%, respectively. The radar chart clearly shows that the TAFE-CLIP model exhibits almost no significant fluctuations in the tail categories, whereas other models, Particularly ResNet-101, demonstrate considerable instability in performance for these categories. For example, in the baseball diamond classification accuracy, the classification accuracy of TAFE-CLIP was greater than 98%, whereas that of ResNet-101 was less than 75%. Although focal loss and CLIP (ViT-B/16) performed well in some classification accuracy, such as tennis court, their performances in other tail categories were still inferior to those of TAFE-CLIP, exhibiting greater variability.

In the RSI-CB256 dataset, TAFE-CLIP achieved a classification accuracy of 97.1% for tail classification accuracy. In the desert and meadow classification accuracy, TAFE-CLIP performed particularly well, with accuracy rates exceeding 96%, whereas the accuracies of ResNet-101 were approximately 63.51% and 70%, respectively. The radar chart illustrates

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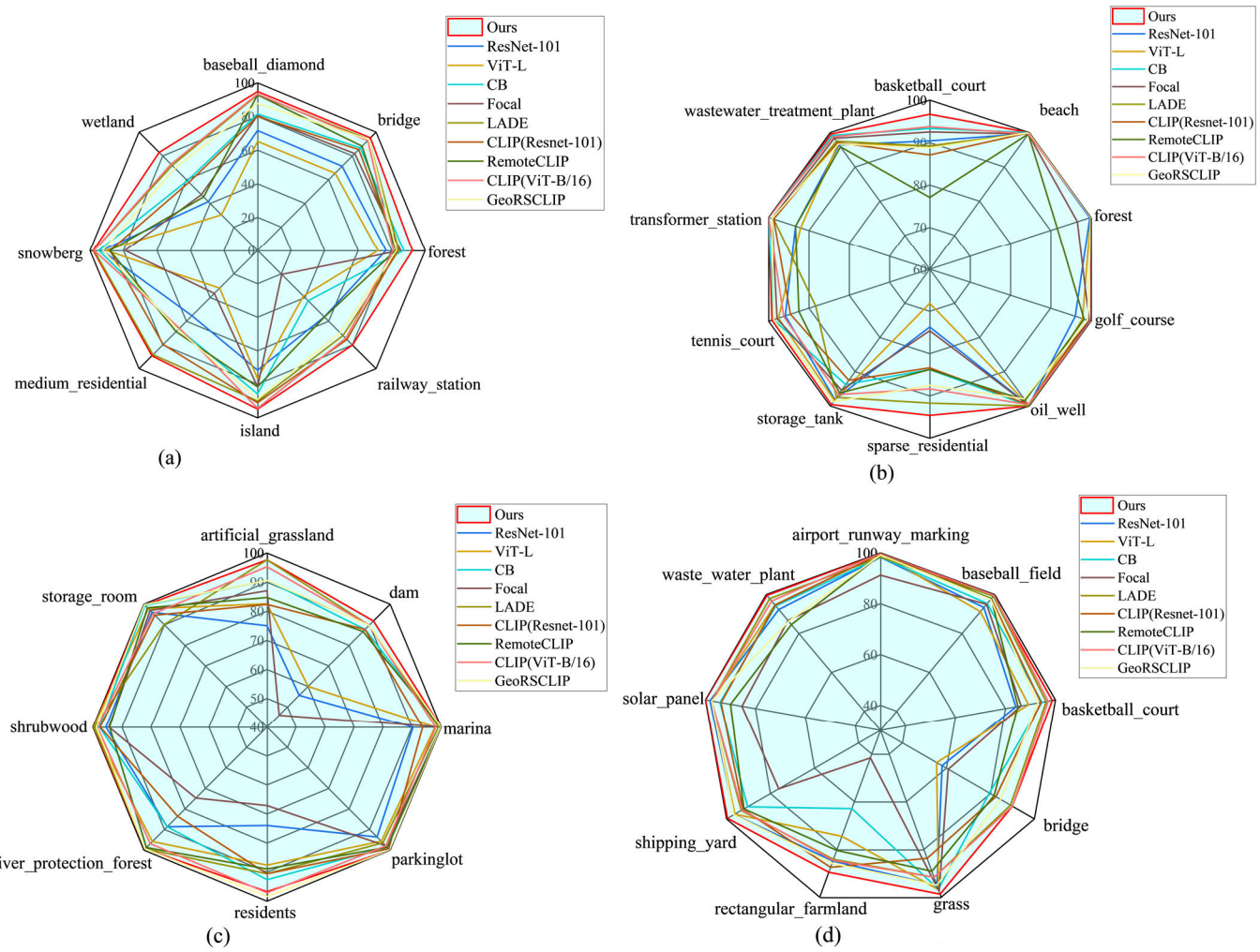


Fig. 7. Comparison of tail classes accuracy across different models in each dataset. (a) NWPU-RESISC45, (b) PatternNet, (c) RSSDIVCS, and (d) RSI-CB256.

the consistency and stability of TAFE-CLIP on this dataset, particularly its nearly perfect performance in tail classes with minimal curve fluctuation. In contrast, ViT-L performed less evenly in the tail classification accuracy, with accuracy in the desert classification accuracy falling below 80%. In comparison, focal loss and CLIP (ViT-B/16) improved their handling of tail categories; however, in some categories, such as meadows, they still trailed behind TAFE-CLIP, with accuracy rates below 90%.

In the RSSDIVCS dataset, TAFE-CLIP achieved a classification accuracy of 91.7% for tail categories. For example, in the runway and stadium categories, TAFE-CLIP's accuracy exceeded 90%, whereas that of ResNet-101 was only about 70.39% and 68%, respectively. The radar chart shows that TAFE-CLIP maintained a clear advantage across multiple tail categories in the RSSDIVCS dataset, particularly in the solar panel category, where the TAFE-CLIP's accuracy was significantly higher than the other models. In contrast, the curves for ResNet-101 and ViT-L showed significant fluctuations.

E. Ablation Study

Ablation studies were performed on four distinct datasets to evaluate how the TAFE module promotes overall classification performance. The model configurations for the comparative analysis are as follows: the baseline configuration excluded both the LA module and the TAFE module. This baseline was compared against configurations using only the LA module, used only the SE channel attention module, used only the CE module, and fully utilized the TAFE module to determine the contribution of each module to OA, $F1$ score, and accuracy results for head, medium, and tail categories. The results of ablation studies and comparative experiments are provided in Tables XI–XIV.

In NWPU-RESISC45 experiments (Table XI), the baseline model (without the TAFE module) achieved 88.5% tail-class classification accuracy. With the TAFE module, this accuracy increased to 90.5%, showing a 2% improvement, highlighting the TAFE module's advantage in handling tail classes.

In the PatternNet dataset (Table XII), the model with the TAFE module achieved 97.5% accuracy for tail categories compared to 82.9% for the baseline, indicating sustained excellent performance in high-accuracy scenarios.

In the RSSDIVCS dataset (Table XIII), the TAFE module's impact was more pronounced. By incorporating the CE module, the accuracy rate increased from 88.5% to 91.7%, an increase of 3.2%, demonstrating its powerful role in enhancing fine-grained feature extraction.

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TABLE XI
EXPERIMENTAL RESULTS OF VARIOUS MODULES ON NWPU-RESISC45 (%)

Modules	LA	CE	SE	OA (%)	F1 (%)	Head (%)	Medium (%)	Tail (%)
Baseline				77.4	74.6	77.4	94.1	52.2
+ LA	√			83.7	83.6	90.9	85.1	81.5
+ SE	√	√		86.9	86.9	86.8	84.6	88.5
+ CE	√		√	92.7	92.7	96.5	93.8	89.7
Ours	√	√	√	**93.2**	**93.1**	**95.3**	**94.4**	**90.5**

TABLE XII
EXPERIMENTAL RESULTS OF VARIOUS MODULES ON PATTERNNET (%)

Modules	LA	CE	SE	OA (%)	F1 (%)	Head (%)	Medium (%)	Tail (%)
Baseline				92.4	92.1	99.3	99.3	82.9
+ LA	√			96.9	96.9	95.1	97.0	97.5
+ SE	√	√		96.0	96.1	92.0	96.7	97.1
+ CE	√		√	99.1	99.10	99.2	99.3	98.8
Ours	√	√	√	**99.3**	**99.3**	**99.5**	**99.7**	**98.8**

TABLE XIII
EXPERIMENTAL RESULTS OF VARIOUS MODULES ON RSSDIVCS (%)

Modules	LA	CE	SE	OA (%)	F1 (%)	Head (%)	Medium (%)	Tail (%)
Baseline				79.1	77.0	95.8	96.7	52.5
+ LA	√			91.4	91.4	89.8	93.0	90.6
+ SE	√	√		89.3	89.3	87.5	90.9	88.5
+ CE	√		√	94.4	94.3	97.3	96.9	90.7
Ours	√	√	√	**95.2**	**95.1**	**97.9**	**97.3**	**91.7**

TABLE XIV
EXPERIMENTAL RESULTS OF VARIOUS MODULES ON RSI-CB256 (%)

Modules	LA	CE	SE	OA (%)	F1 (%)	Head (%)	Medium (%)	Tail (%)
Baseline				83.8	83.3	98.8	97.5	66.0
+ LA	√			91.4	91.4	92.3	94.6	95.5
+ SE	√	√		89.3	89.3	92.5	89.5	93.8
+ CE	√		√	97.8	97.6	99.3	98.1	96.8
Ours	√	√	√	**98.1**	**97.9**	**99.0**	**98.6**	**97.1**

The RSI-CB256 dataset (Table XIV) further validated the TAFE module's effectiveness; compared with the baseline with only LA loss, the model equipped with both the TAFE module and LA loss improves accuracy from 95.5% to 97.1%, achieving a significant 1.6% improvement on a complex and imbalanced dataset. In comparison, the model using only the SE module showed some improvements in tail classification accuracy classification across all datasets; however, the extent of the improvement was limited. For example, in the RSI-CB256 dataset, the classification accuracy for tail classification accuracy reached only 93.8%, which further illustrates that while the SE module enhances channel attention, it lacks fine-grained feature extraction classification accuracy for convolutional enhancement.

Four experimental results demonstrate that the TAFE model can improve classification accuracy based on the distribution characteristics of remote sensing long-tail data. By adding the LA model and comparing it with the baseline model, the accuracy of tail-class samples has been significantly improved. Although the SE module can improve classification accuracy, only the combination of CE and channel attention mechanism can synchronously enhance the model's ability to extract fine-grained texture features of features and realize the modeling of the global spatial dependence of remote sensing scene. The results fully confirm that the TAFE module can improve the generalization ability of the TAFE-CLIP model and has significant advantages in solving the problems of imbalanced categories and complex distribution of features in remote sensing image classification tasks.

F. Comparison Between Multimodal and Unimodal Models in Classification Tasks

Low recognition accuracy of tail classes in long-tailed remote sensing tasks mainly stems from insufficient tail sample quantities and fine-grained similarities among diverse categories. By analyzing the confusion matrices of the TAFE-CLIP, CLIP (ResNet-101), ResNet-101, and ViT-L models, as shown in Fig. 8, it can be found that the multimodal approach had remarkable advantages in identifying rare tail samples and reducing misclassification between categories. The core purpose of this research on the CLIP model of ResNet-101 was to verify the performance improvement effect of the CE channel attention module TAFE in this article. The CLIP model with ResNet-101 as the visual encoder is set as the baseline model, which can more clearly reflect the performance gain brought by the CE branch in the long-tail feature extraction task. Section III-F analyzed the category-specific classification accuracy from confusion matrix outcomes, discussed advantages and limitations of each model, and evaluated their performance in fine-grained feature discrimination tasks.

Considering the comprehensive results from the confusion matrix, the unimodal models (ResNet-101 and ViT-L) generally exhibited lower accuracy, particularly in visually similar classes, where they struggled to distinguish fine-grained differences, leading to significant confusion between multiple similar classes, as shown in Fig. 9. For instance, ResNet-101 achieved only 32.9% accuracy in distinguishing artificial grassland from green farmland, frequently confusing them with other similar categories like avenues. ViT-L performed slightly better at 68.2% but still showed limitations in extracting fine-grained features. In natural scenes, such as river protection forest and river, ResNet-101 recorded a low accuracy of 29.6%, often misclassifying rivers, while ViT-L reached 37.8%, indicating persistent challenges. Similar issues arise between dams and bridges, where ResNet-101 and ViT-L achieved only 29.6% and 37.8% accuracy, respectively, often mistaking one for the other.

In contrast, the multimodal CLIP model based on ResNet-101 backbone network achieved better classification results by fusing remote sensing image features and semantic information of land objects. For example, CLIP achieved 82.4% accuracy in artificial grassland classification and 83.5% for river protection forest, notably reducing confusion compared to unimodal models, although some misclassification remained. In the dam classification task, the CLIP framework

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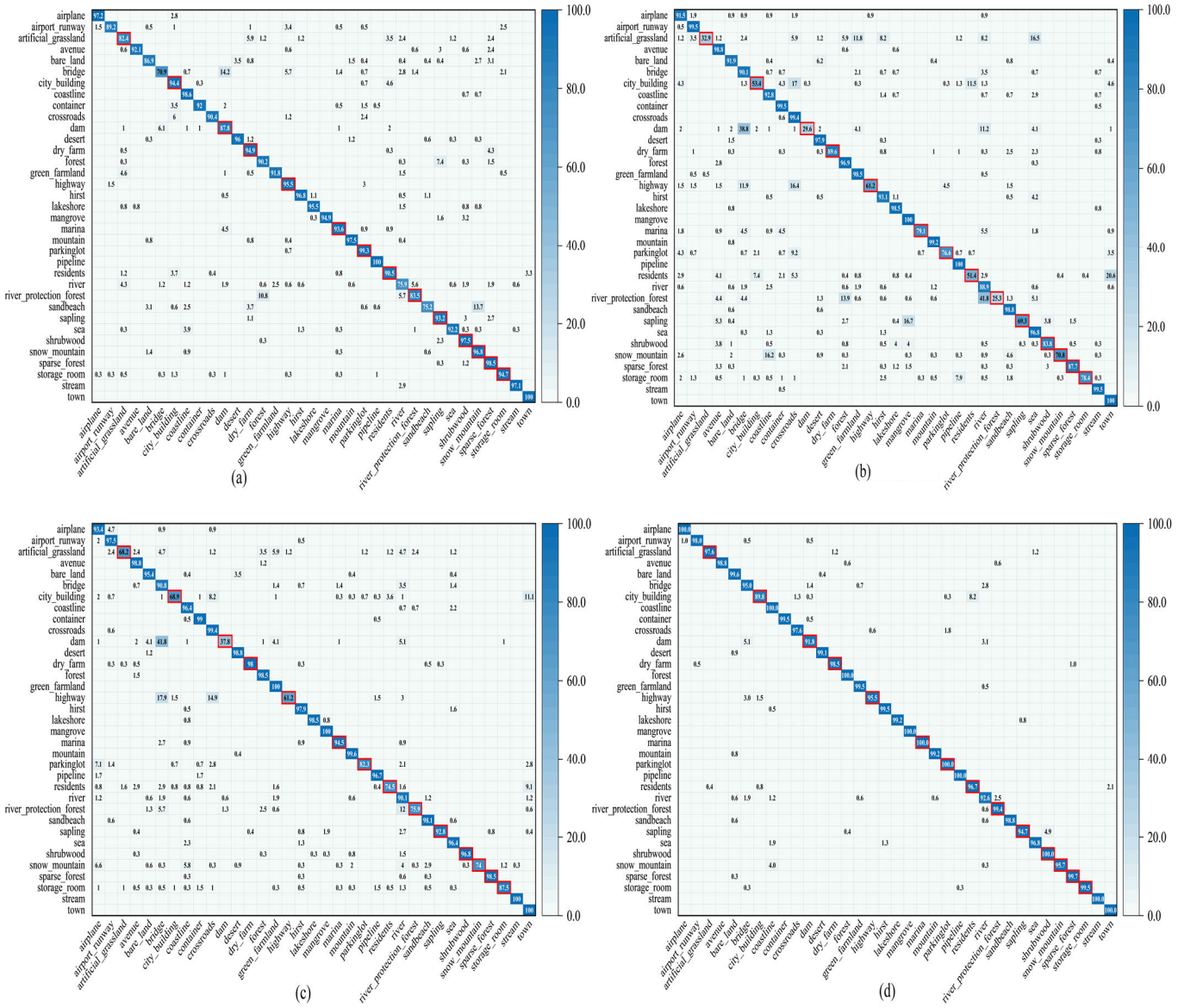


Fig. 8. Confusion matrix of RSI-CB256 across different models. (a) CLIP(Resnet-101). (b) Resnet-101. (c) Vit-L. (d) Ours.

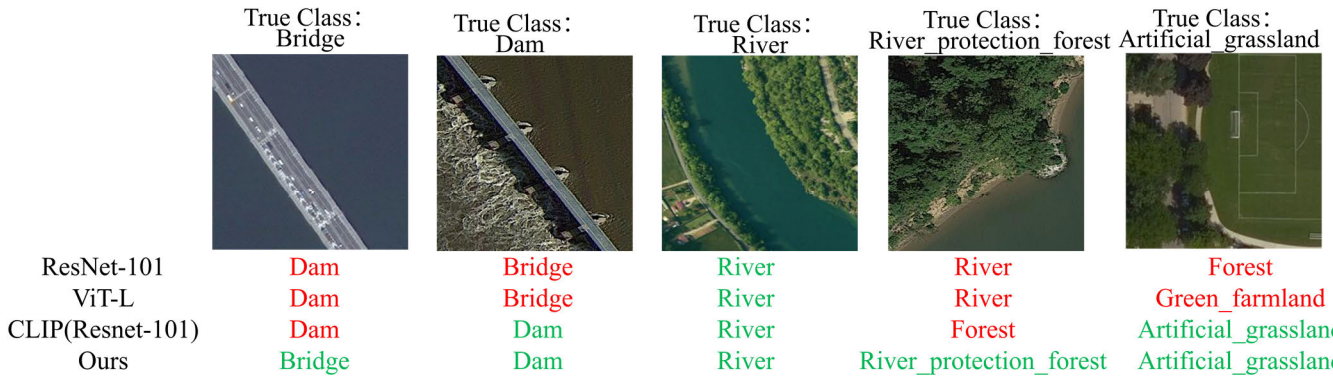


Fig. 9. Prediction results of different models. Incorrect forecasts are highlighted in red, and correct ones are shown in green.

achieved a classification accuracy of 87.8%, which outperformed the ResNet-101 model by a notable margin; however, the issue of visual similarity between dams and bridges has not yet been effectively resolved and remains a major

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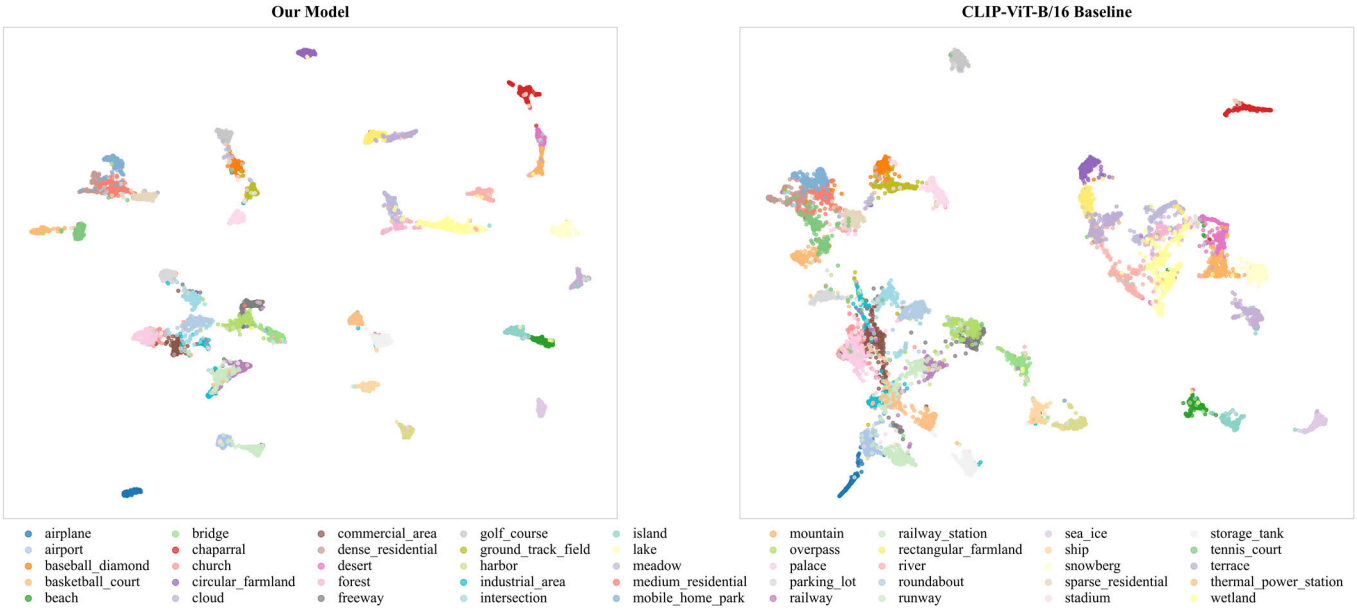


Fig. 10. NWPU-RESISC45 dataset, UMAP feature visualization diagram of TAFE-CLIP method (left) and baseline method (right).

challenge in current classification tasks. By introducing CE and combining it with multimodal information, the TAFE-CLIP model further improved accuracy. As illustrated in Fig. 8, in river protection forest classification, TAFE-CLIP achieved 96.7%, and in artificial grassland classification, it reached 97.6%, surpassing both CLIP and ViT-L. Most notably, in dam classification, TAFE-CLIP improved accuracy to 91.8%, significantly reducing confusion between dams and bridges, and proves that the model can effectively extract the fine-grained features of visual similar features in remote sensing images.

Overall, the classification accuracy of unimodal models for long-tail ground objects was much lower than that of multi-mode framework. In visually similar categories, the unimodal models were prone to classification confusion, which exposes obvious performance defects. Multimodal methods improved classification effect and reduced misclassification between categories by integrating visual and text semantic information. Feature embeddings of TAFE-CLIP and baseline CLIP were visualized on NWPU-RESISC45 using UMAP (Fig. 10). Three key observations confirm our method's effectiveness: tighter clustering for tail classes, confirming effective few-shot learning; distinct separation of visually similar categories; and better semantic alignment reflecting linguistic knowledge. These results empirically validate that TAFE-CLIP improves both discriminability and semantic coherence. TAFE-CLIP has powerful local feature extraction and global semantic understanding capabilities. Therefore, the multimodal architecture represented by TAFE-CLIP is very suitable for the long-tail remote sensing image classification task.

G. Comparative Analysis of the TAFE Module and Typical Attention Structures

To more comprehensively validate the validity of the TAFE module, this article conducts in-depth comparative evaluations against mainstream attention modules, including CA [60], CBAM [61], ECA [48], EMA [62], and ELA [63]. As shown in Tables XV–XVIII, these tables present the performance on different datasets. Experimental findings revealed that the TAFE module obviously surpassed other attention mechanisms in terms of overall accuracy and $F1$ scores. Especially in classification tasks involving long-tail categories, the TAFE module demonstrated stronger fine-grained feature extraction capabilities, effectively mitigating the challenges posed by long-tail distributions.

TABLE XV
ACCURACY OF VARIOUS ATTENTION MECHANISMS ON NWPU-RESISC45

Attention	OA (%)	F1(%)	Head (%)	Medium (%)	Tail (%)
CA	92.1	92.1	96.0	93.3	89.0
CBAM	92.1	92.1	95.8	93.4	88.9
ECA	93.0	93.0	96.8	93.9	90.2
ELA	91.5	91.5	95.6	92.3	88.7
EMA	92.7	92.7	96.5	93.8	89.7
TAFE	**93.2**	**93.1**	**95.3**	**94.4**	**90.5**

According to Table XV, the ECA model accuracy was closest to that of the TAFE module, with both achieving an overall accuracy and $F1$ metric reaching 93.0%. The CA and CBAM modules both reached 92.1% in both accuracy and $F1$ value, while the ELA module performed poorly, attaining 91.5% for both accuracy and $F1$ metric. In terms of tail-class classification, the TAFE module outperformed the other comparison modules, achieving a tail-class accuracy of 90.5%. The ECA module ranked second with an accuracy of 90.2%, followed closely by CA and CBAM at 89% and 88.9%, respectively, while the ELA module had the lowest accuracy at 88.7%. Notably, the EMA module achieved an accuracy of 89.7% on this task, with performance falling between that of CA, CBAM, and ECA. This significant performance

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TABLE XVI
ACCURACY OF VARIOUS ATTENTION MECHANISMS ON PATTERNNET

Attention	OA (%)	F1(%)	Head (%)	Medium (%)	Tail (%)
CA	98.9	98.9	99.5	99.4	98.1
CBAM	98.9	98.9	99.5	99.5	98.1
ECA	99.2	99.2	99.3	99.5	98.8
ELA	98.8	98.8	99.3	99.3	98.2
EMA	99.1	99.1	99.2	99.3	98.8
TAFE	**99.3**	**99.3**	**99.5**	**99.7**	**98.8**

TABLE XVII
ACCURACY OF VARIOUS ATTENTION MECHANISMS ON RSSDIVCS

Attention	OA (%)	F1(%)	Head (%)	Medium (%)	Tail (%)
CA	94.3	94.4	97.6	96.9	90.6
CBAM	94.3	94.2	97.4	96.6	90.8
ECA	94.4	94.3	97.1	96.8	90.8
ELA	93.9	94.0	97.1	96.5	90.2
EMA	94.3	94.4	97.3	96.9	90.7
TAFE	**95.2**	**95.1**	**97.9**	**97.3**	**91.7**

improvement fully demonstrated that the TAFE module possessed exceptional fine-grained feature extraction capabilities when handling rare classes. In contrast, while other attention mechanisms achieved acceptable results for head classes, their performance dropped significantly for tail classes.

Further analysis showed that the TAFE module demonstrated significant advantages in recognizing tail-class samples while maintaining stable classification performance for head and middle samples. On the NWPU-RESISC45 dataset, the TAFE module reached a head-class accuracy of 95.3% and a midclass accuracy of 94.4%. Although the ECA module slightly outperformed the TAFE module in head-class accuracy with 96.8%, the TAFE module demonstrated greater stability in midclass accuracy, where the ECA module achieved 93.9%. The ELA module reached a head-category accuracy of 95.6% and a midcategory accuracy of 92.3%; the EMA module reached a head-category accuracy of 96.5% and a midcategory accuracy of 93.8%. These results fully demonstrated the TAFE module's balance and stability across different categories.

From the outcomes in Table XVI, we found that the TAFE module performed exceptionally well on the PatternNet dataset. The ECA module followed closely behind both the overall accuracy, and the $F1$ score reached 99.2%; the CA and CBAM modules both reached 98.9% in overall accuracy with matching $F1$ score standing at 98.9%; while the ELA module performs poorly, securing accuracy at 98.8% coupled with $F1$ value hitting 98.8%. In the tail-class classification task, the TAFE module also achieved remarkable efficacy and reached a tail-class accuracy of 98.8%, significantly outperforming the other comparison modules. The ECA module also achieved a tail-class accuracy of 98.8% in this task, performing on par with the TAFE module; the CA and CBAM modules both achieved a tail-class accuracy of 98.1%, while the ELA module performed the weakest with an accuracy of 98.2%. Notably, the EMA module also achieved an accuracy of 98.8% in this task. This result further validates that the TAFE module possesses stronger fine-grained feature extraction capabilities when handling rare classes, whereas other attention mechanisms, although performing well on head classes, show relatively weaker performance on tail classes. Although the ECA module slightly outperformed the TAFE module on head categories with an accuracy of 99.3%, the TAFE module demonstrated greater stability in middle categories, while the ECA module achieved an accuracy of 99.5% on middle categories. The performance of other modules is as follows: CA and CBAM both achieve 99.5% accuracy on head categories, with middle category accuracies of 99.4% and 99.5%, respectively; the ELA module achieved a top-category accuracy of 99.3% and a middle-category accuracy of 99.3%. These results fully demonstrated the TAFE module's balance and robustness across different categories, further validated its advantages in practical applications.

Based on the performance data summarized in Table XVII, the TAFE module demonstrated a significant performance advantage on the RSSDIVCS dataset. The ECA module performed similar to the TAFE module, achieving 94.4% OA paired with 94.3% $F1$ value; The CA, CBAM, and EMA modules all achieved an overall accuracy of 94.3%, with $F1$ values are 94.4%, 94.2%, and 94.4% in sequence; in contrast, the ELA module performed relatively poorly, with an overall accuracy of 93.9% together with a 94.0% $F1$ metric.

In the tail-class classification task, the TAFE module performed particularly well, achieving a tail-class accuracy of 91.7%, significantly higher than that of other comparison modules. The ECA and CBAM modules both achieved an accuracy of 90.8% in this task, performing similar to the TAFE module; the CA and EMA modules achieved tail classes accuracy rates of 90.6% and 90.7%, respectively; meanwhile, the ELA module performed the weakest with an accuracy rate of 90.2%. This significant performance improvement indicated that the TAFE module possessed stronger fine-grained feature extraction capabilities when handling rare categories. In contrast, while other attention mechanisms CA and CBAM performed well in the head categories, their performance on tail categories was weak. Although the CA module slightly outperforms the TAFE module on head categories with an accuracy of 97.6%, the TAFE module demonstrates more stable performance in middle categories, where CA achieves an accuracy of 96.9%. The performance of other modules is as follows: CBAM and EMA achieved head-of-class accuracies of 97.4% and 97.3%, respectively, and mid-of-class accuracies of 96.6% and 96.9%, respectively. These results clearly demonstrate that the TAFE module can effectively balance classification tasks for common and rare classes, showcased its strong adaptability and robustness in complex scenarios.

The experimental data in Table XVIII clearly demonstrated the outstanding performance of the TAFE module on the RSI-CB256 dataset. The ECA module performed similar to the TAFE module, achieving an overall accuracy reaching 97.7% and $F1$ metric is 97.8%; The CA and EMA modules both achieved an overall accuracy reaching 97.6% and $F1$ metric is 97.5% and 97.8%, respectively; in contrast, the CBAM module performed relatively poorly, reaching 97.5% overall accuracy and 97.2% $F1$ metric. In the classification

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TABLE XVIII
ACCURACY OF VARIOUS ATTENTION MECHANISMS ON RSI-CB256

Attention	OA (%)	F1(%)	Head (%)	Medium (%)	Tail (%)
CA	97.6	97.5	98.9	98.0	96.5
CBAM	97.5	97.2	99.1	98.0	95.8
ECA	97.7	97.8	99.2	98.3	96.6
ELA	97.5	97.8	99.2	97.9	96.9
EMA	97.6	97.8	99.3	98.1	96.8
TAFE	**98.1**	**97.9**	**99.0**	**98.6**	**97.1**

task for tail categories, the TAFE module performed particularly well, achieving a tail classes accuracy of 97.1%, significantly higher than the other comparison modules. The ECA module achieved an accuracy of 96.6% in this task, performing similar to the TAFE module; the ELA module achieved a tail-class accuracy of 96.9%, also performing well; the EMA module achieved a tail-class accuracy of 96.8%; while the CA and CBAM modules achieved tail-class accuracies of 96.5% and 95.8%, respectively, performing relatively weakly. These results further validated that the TAFE module possessed stronger fine-grained feature extraction capabilities when handling tail categories. Although the ECA module slightly outperforms the TAFE module in top-tier categories with an accuracy of 99.2%, the TAFE module demonstrates more stable performance in midtier categories, where the ECA module achieves an accuracy of 98.3%. The performance of other modules is as follows: CA and EMA achieved head-class accuracies of 98.9% and 99.3%, respectively, and midclass accuracies of 98.0% and 98.1%, respectively; the CBAM module achieved 99.1% accuracy on head categories and 98.0% for middle categories. These results clearly demonstrated that the TAFE module was capable of comprehensively handling classification tasks across top, middle, and bottom categories, showcased its strong adaptability and robustness in complex remote sensing image classification scenarios.

The analysis of the experimental results across four datasets shows that the TAFE module demonstrates comprehensive and significant advantages when compared to various attention mechanisms. In terms of overall performance, the TAFE module achieved higher overall accuracy and macro-average $F1$ scores than other compared modules across all four datasets. Its performance was particularly outstanding in classification tasks involving tail categories, where its accuracy for these categories significantly outperformed other modules. This indicated that the TAFE module possessed stronger fine-grained feature extraction capabilities when handling rare categories. Furthermore, the TAFE module maintained high-precision classification performance for head and middle categories, demonstrated its balance and stability between common and rare categories. The TAFE module small receptive field of these convolutions is ideally suited for distinguishing fine-grained tail-class patterns without introducing background noise, while the residual shortcut preserves scarce tail-class information from being diluted. In contrast, CA and ELA emphasize long-range dependencies that can overwhelm weak tail signals, and CBAM's max pooling discards critical edge details. The TAFE module's comprehensive advantages across multiple datasets fully demonstrated its strong adaptability and robustness in complex scenarios.

V. CONCLUSION

To address the class imbalance and feature ambiguity challenges of long-tail distributions in remote sensing image classification, this article proposes the TAFE-CLIP model. The model addresses long-tailed remote sensing classification by integrating a TAFE module into CLIP for enhanced tail-class feature learning. Paired with a cosine classifier and LA loss, it boosts tail-class precision without sacrificing head/medium-class performance, filling the gap of few-shot adaptation in current CLIP models. The experimental findings revealed that the TAFE-CLIP model attained tail classification accuracies of 90.5%, 98.8%, 91.7%, and 97.1% across four datasets (NWPU-RESISC45, PatternNet, RSSDIVCS, and RSI-CB256). Compared with four widely adopted deep learning baselines, including ResNet-101, EfficientNet, ViT-L, and Swin Transformer, the proposed model improved the overall average classification accuracy by 45.1%, 15.03%, 18.92%, and 18.51%. Furthermore, TAFE-CLIP yields tail-class accuracy improvements of 11.2%, 3.7%, 5.7%, and 10.73% across the four datasets, respectively.

Ablation studies further validated the effectiveness of the proposed TAFE module. Compared with the standard CLIP (ViT-B/16) baseline without the TAFE module, the TAFE-enhanced framework achieved an average 30% performance gain on minority classes. Notably, the TAFE-CLIP framework consistently improved classification accuracy for both head and tail classes, demonstrating its efficacy in mitigating long-tail bias under complex remote sensing scene conditions.

TAFE is a fixed structure whose convolutional receptive field cannot adaptively adjust to the scales of input ground objects. When applied to long-tail datasets with huge differences in scale distribution, the limitations of the existing attention and feature enhancement schemes will become more pronounced. Future studies will investigate more advanced multimodal learning approaches and integrate a scale-adaptive weight allocation module to further enhance the adaptability and overall performance of the framework in multiscale tail categories.

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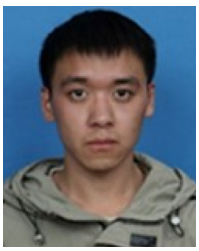
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